Type MS · Finding stationary points

Answer 1 of 8 · Question 7 part a · 5 marks

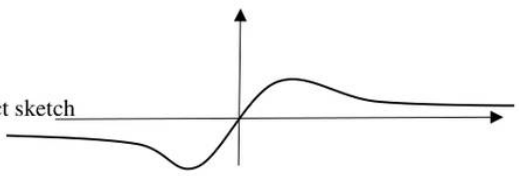
2021 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

7 (a)	$\frac{dy}{dx} = \frac{(x^2+4)-x(2x)}{(x^2+4)^2}$	M1
	$= \frac{4-x^2}{(x^2+4)^2}$	A1
	$\frac{dy}{dx} = 0 \Rightarrow 4-x^2 = 0$	M1
	So $\left(2, \frac{1}{4}\right)$ and $\left(-2, -\frac{1}{4}\right)$	A1 A1 (5)
(b)	$\frac{d^2y}{dx^2} = \frac{-2x(x^2+4)^2 - 4x(4-x^2)(x^2+4)}{(x^2+4)^4}$	M1
	$\frac{d^2y}{dx^2} = \frac{-2x^3-8x-16x+4x^3}{(x^2+4)^3}$	M1
	$\frac{d^2y}{dx^2} = \frac{2x^3-24x}{(x^2+4)^3}$	M1
	$\frac{d^2y}{dx^2} = \frac{2x(x^2-12)}{(x^2+4)^3}$ *	A1 cso (4)
(c)	When $x = 2$ $\left[\frac{d^2y}{dx^2} = -\frac{1}{16}\right]$ When $x = -2$ $\left[\frac{d^2y}{dx^2} = \frac{1}{16}\right]$	
	$\frac{d^2y}{dx^2} < 0$ so maximum $\frac{d^2y}{dx^2} > 0$ so minimum	M1 A1ft (2)
		Total 11 marks

Type MS · Finding stationary points

Answer 1 of 8 · Question 7 part a · 5 marks

2021 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

	A1	For the correct coordinates of either $\left(2,\frac{1}{4}\right)$ or $\left(-2,-\frac{1}{4}\right)$
	A1	For both correct coordinates $\left(2,\frac{1}{4}\right)$ and $\left(-2,-\frac{1}{4}\right)$
(b)	**M1**	For an attempt at Quotient rule on their $\frac{dy}{dx}=\frac{(x^2+4)-x(2x)}{(x^2+4)^2}$ which must be as a minimum: $\frac{ax^2+bx+c}{(x^2+4)^2}$ where a, b and c are constants and $a,c\neq 0$ The definition of an attempt is that there must be a minimal attempt to differentiate the numerator and denominator and the correct formula applied. $(x^2+4)^2$ must differentiate to $ax(x^2+4)$, the denominator must be $(x^2+4)^4$ Allow the terms in the numerator to be the wrong way around, but the terms must be subtracted. Apply General Guidance for an attempt at differentiation on ax^2+bx+c. $\frac{d^2y}{dx^2}=\frac{-2x(x^2+4)^2-4x(4-x^2)(x^2+4)}{(x^2+4)^4}$
	M1	For cancelling through by (x^2+4) $\frac{d^2y}{dx^2}=\frac{-2x(x^2+4)-4x(4-x^2)}{(x^2+4)^3}$
	M1	For simplifying the numerator to achieve as a minimum $\frac{d^2y}{dx^2}=\frac{ax^3+bx}{(x^2+4)^3}$ where a and b are constants
	A1 cso	For obtaining the answer as given with no errors. $\frac{d^2y}{dx^2}=\frac{2x(x^2-12)}{(x^2+4)^3}$ *
(c)	**M1**	Substitutes either their 2 or their – 2 into their $\frac{d^2y}{dx^2}$ Note: When $x=2\left[\frac{d^2y}{dx^2}=-\frac{1}{16}\right]$ and when $x=-2\left[\frac{d^2y}{dx^2}=\frac{1}{16}\right]$
	A1ft	For the conclusion $\frac{d^2y}{dx^2}<0$ so maximum $\frac{d^2y}{dx^2}>0$ so minimum
	ALT – tests gradient or sight of a sketch	
	M1	Tests gradient on either side of one the turning points (their 2 or their – 2) using their $\frac{dy}{dx}$ or a correct sketch 
	A1ft	For the correct conclusion

Type MS · Finding stationary points

Answer 1 of 8 · Question 7 part a · 5 marks

2021 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding stationary points

Answer 2 of 8 · Question 9 part b.ii · 7 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part b.ii in this mark scheme extract

9(a)	$f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\ (+c)$	M1A1
	$x=-2\ \ y=-\frac{28}{3}\ \Rightarrow c=0$	M1
	$\left(f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\right)\ \therefore C$ passes through O	A1 cso (4)
(b)(i)	$x=2\ \ f'(x)=8-4-8+4=0$	M1
	$\frac{d^2y}{dx^2}=3x^2-2x-4$	M1
	$x=2\ \ \frac{d^2y}{dx^2}=12-4-4>0\ \ \therefore$ min at $x=2$	A1cso
	$x=1\ \ f'(x)=1-1-4+4=0$	M1
	$x=1\ \ \frac{d^2y}{dx^2}=3-2-4<0\ \ \therefore$ max at $x=1$	A1 cso
ALT	$f'(x)=(x-2)(x-1)(x+2)\ (=0)$ factorise	M1
	$x=2,1,(-2)$ solve (solutions to be 2,1 (and another))	M1
	OR: $f'(x)\ (=0)$ solved by calculator.	
	All 3 solutions needed (and correct) $=0$ not needed M2	
	$\frac{d^2y}{dx^2}=3x^2-2x-4$ differentiate	M1
	$x=2\ \ \frac{d^2y}{dx^2}=12-4-4>0\ \ \therefore$ min at $x=2$	A1cso
	$x=1\ \ \frac{d^2y}{dx^2}=3-2-4<0\ \ \therefore$ max at $x=1$	A1cso
(ii)	$x=1\Rightarrow y=1\frac{11}{12}\qquad x=2\Rightarrow y=1\frac{1}{3}$	B1B1 (7)
(c)	$y'=(x-1)(x-2)(x+2)$	M1
(i)	$x=-2,\ y=-\frac{28}{3}$ or $\left(-2,-\frac{28}{3}\right)$	A1
(ii)	$x=-2\ \ \frac{d^2y}{dx^2}=12+4-4>0\ \ \therefore$ min point	A1cso (3)
		[14]

Type MS · Finding stationary points

Answer 2 of 8 · Question 9 part b.ii · 7 marks

2019 JUN · Paper 02 · extract 2 of 3 · find part b.ii in this mark scheme extract

Question Number	Scheme	Marks
(a)M1	Attempt to integrate f'(x). The power of at least one x term must increase and none should decrease. c not needed	
A1	Correct integration, c not needed	
M1	Substitute the given coordinates to show $c = 0$. If c is not included (or assumed to be 0), then showing that substitution of $x = -2$ gives $y = -28/3$ is acceptable. Substitutions must be shown.	
A1cso	Correct conclusion from fully correct work. Accept eg $f(0) = 0$ ∴ shown	
(b)	Ignore labels (i) and (ii) when marking (b)	
(i)M1	Substitute $x = 2$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
M1	Differentiate the expression for f'(x). At least one power must decrease and none increase.	
A1cso	Show second derivative is > 0 at $x = 2$ and give the conclusion. No errors or omissions in the working.	
M1	Substitute $x = 1$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
A1cso	Show second derivative is < 0 at $x = 1$ and give the conclusion. No errors or omissions in the working.	
(ii)B1	For either y coordinate correct (and x coordinate correctly indicated; substitution shown indicates this)	
B1	For the second y coordinate correct	
(c)	(May have been seen in (b))	
M1	Factorise f'(x) completely – any valid method OR use the factor theorem to find $x = -2$	
(i)A1	Extract the x coordinate of the third turning point and obtain the corresponding y coordinate. May quote y coordinate from the question	
(ii)A1cso	Test the sign of the second derivative at this point and make the conclusion. All work in (c) and $\frac{d^2y}{dx^2}$ (from (b)) must be completely correct for this mark to be awarded.	
	Alternative ways to determine the nature of the turning points:	
1.	If the change of sign of f'(x) is used then values of f'(x) either side of 1 and 2 must be calculated to provide evidence.	
2.	The continuity of a cubic function can be used to establish the nature of the turning points. If in doubt send to review.	

Type MS · Finding stationary points

Answer 2 of 8 · Question 9 part b.ii · 7 marks

2019 JUN · Paper 02 · extract 3 of 3 · find part b.ii in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Finding stationary points

Answer 3 of 8 · Question 9 part c.i · 0 marks

2019 JUN · Paper 02 · extract 1 of 3 · find part c.i in this mark scheme extract

9(a)	$f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\ (+c)$	M1A1
	$x=-2\ \ y=-\frac{28}{3}\ \Rightarrow c=0$	M1
	$\left(f(x)=\frac{1}{4}x^4-\frac{1}{3}x^3-2x^2+4x\right)\ \therefore C$ passes through O	A1 cso (4)
(b)(i)	$x=2\ \ f'(x)=8-4-8+4=0$	M1
	$\frac{d^2y}{dx^2}=3x^2-2x-4$	M1
	$x=2\ \ \frac{d^2y}{dx^2}=12-4-4>0\ \ \therefore$ min at $x=2$	A1cso
	$x=1\ \ f'(x)=1-1-4+4=0$	M1
	$x=1\ \ \frac{d^2y}{dx^2}=3-2-4<0\ \ \therefore$ max at $x=1$	A1 cso
ALT	$f'(x)=(x-2)(x-1)(x+2)\ (=0)$ factorise	M1
	$x=2,1,(-2)$ solve (solutions to be 2,1 (and another))	M1
	OR: $f'(x)\ (=0)$ solved by calculator.	
	All 3 solutions needed (and correct) $=0$ not needed M2	
	$\frac{d^2y}{dx^2}=3x^2-2x-4$ differentiate	M1
	$x=2\ \ \frac{d^2y}{dx^2}=12-4-4>0\ \ \therefore$ min at $x=2$	A1cso
	$x=1\ \ \frac{d^2y}{dx^2}=3-2-4<0\ \ \therefore$ max at $x=1$	A1cso
(ii)	$x=1\Rightarrow y=1\frac{11}{12}\qquad x=2\Rightarrow y=1\frac{1}{3}$	B1B1 (7)
(c)	$y'=(x-1)(x-2)(x+2)$	M1
(i)	$x=-2,\ y=-\frac{28}{3}$ or $\left(-2,-\frac{28}{3}\right)$	A1
(ii)	$x=-2\ \ \frac{d^2y}{dx^2}=12+4-4>0\ \ \therefore$ min point	A1cso (3)
		[14]

Type MS · Finding stationary points

Answer 3 of 8 · Question 9 part c.i · 0 marks

2019 JUN · Paper 02 · extract 2 of 3 · find part c.i in this mark scheme extract

Question Number	Scheme	Marks
(a)M1	Attempt to integrate f'(x). The power of at least one x term must increase and none should decrease. c not needed	
A1	Correct integration, c not needed	
M1	Substitute the given coordinates to show $c = 0$. If c is not included (or assumed to be 0), then showing that substitution of $x = -2$ gives $y = -28/3$ is acceptable. Substitutions must be shown.	
A1cso	Correct conclusion from fully correct work. Accept eg $f(0) = 0$ ∴ shown	
(b)	Ignore labels (i) and (ii) when marking (b)	
(i)M1	Substitute $x = 2$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
M1	Differentiate the expression for f'(x). At least one power must decrease and none increase.	
A1cso	Show second derivative is > 0 at $x = 2$ and give the conclusion. No errors or omissions in the working.	
M1	Substitute $x = 1$ in the expression for f'(x) to show f'(x) = 0. Substitution must be shown	
A1cso	Show second derivative is < 0 at $x = 1$ and give the conclusion. No errors or omissions in the working.	
(ii)B1	For either y coordinate correct (and x coordinate correctly indicated; substitution shown indicates this)	
B1	For the second y coordinate correct	
(c)	(May have been seen in (b))	
M1	Factorise f'(x) completely – any valid method OR use the factor theorem to find $x = -2$	
(i)A1	Extract the x coordinate of the third turning point and obtain the corresponding y coordinate. May quote y coordinate from the question	
(ii)A1cso	Test the sign of the second derivative at this point and make the conclusion. All work in (c) and $\frac{d^2y}{dx^2}$ (from (b)) must be completely correct for this mark to be awarded.	
	Alternative ways to determine the nature of the turning points:	
1.	If the change of sign of f'(x) is used then values of f'(x) either side of 1 and 2 must be calculated to provide evidence.	
2.	The continuity of a cubic function can be used to establish the nature of the turning points. If in doubt send to review.	

Type MS · Finding stationary points

Answer 3 of 8 · Question 9 part c.i · 0 marks

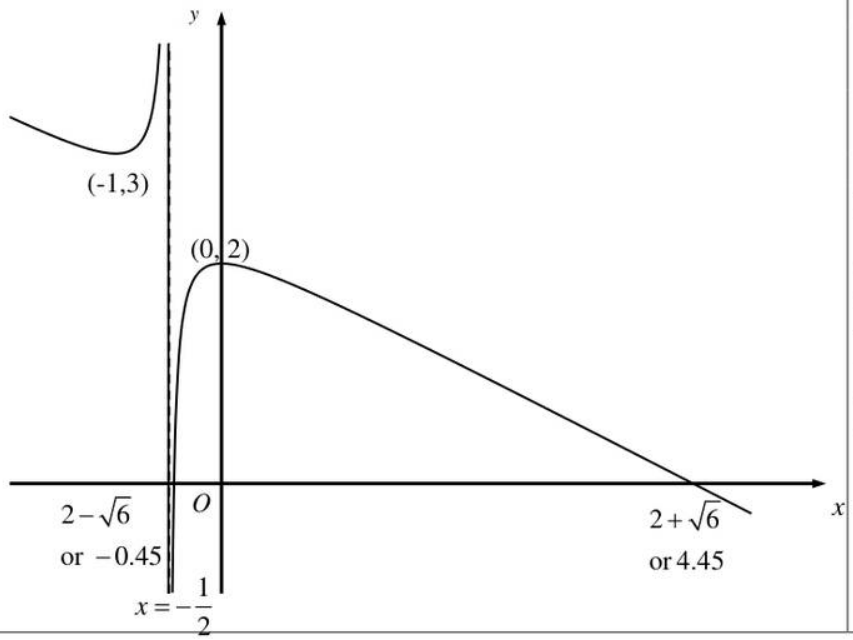
2019 JUN · Paper 02 · extract 3 of 3 · find part c.i in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Finding stationary points

Answer 4 of 8 · Question 9 part c · 6 marks

2020 NOV · Paper 02 · extract 1 of 3 · find part c in this mark scheme extract

9(a)	$y=\dfrac{2+4x-x^2}{2x+1} \Rightarrow x^2-4x-2+2yx+y\ (=0)$ $x^2+(2y-4)x+(y-2)=0$	M1A1A1 (3)
(b)	$(2y-4)^2 \geq 4(y-2)$ $4y^2-16y+16=4y-8 \Rightarrow 4y^2-20y+24=0$ $y \leq 2$ or $y \geq 3$ *	M1 M1A1 A1cso (4)
(c)	$y=\dfrac{2+4x-x^2}{2x+1}$ $\dfrac{dy}{dx}=\dfrac{(4-2x)(2x+1)-2(2+4x-x^2)}{(2x+1)^2}$ **See notes for product rule method** $\dfrac{dy}{dx}=0 \Rightarrow (4-2x)(2x+1)-2(2+4x-x^2)=0$ $2x(x+1)=0 \Rightarrow x=0,\ -1$ stationary points are $(0,2)\ (-1,3)$	M1A1A1 M1 A1 A1 (6)
ALT	$x^2+(2y-4)x+(y-2)=0 \Rightarrow 2x+(2y-4)+2\dfrac{dy}{dx}x+\dfrac{dy}{dx}=0$ $\dfrac{dy}{dx}=0 \Rightarrow x+y=2$ (using (b)) stationary points are $(0,2)\ (-1,3)$	M1A1A1 M1 A1A1
(d)		B1 curve (i) M1A1 (M1 finding coords, A1 correct (oe or min 2dp) and on diagram (ii) B1 (iii) B1ft (5) [18]

Type MS · Finding stationary points

Answer 4 of 8 · Question 9 part c · 6 marks

2020 NOV · Paper 02 · extract 2 of 3 · find part c in this mark scheme extract

(a)M1	Re-write the equation of C without fractions and rearrange to the required form.
A1	Correct value for a and either b or c. These values need not be stated explicitly.
A1	Fully correct equation. Values of a, b and c need not be stated explicitly. Condone missing brackets round "$y-2$". Award this mark when the equation is reached – isw any listing of values with incorrect signs.
(b)	
M1	Use "$b^2 \geqslant 4ac$" for their equation
M1	Re-arrange their inequality to a 3TQ in y. Allow an equation here.
A1	Correct 3TQ, as shown or any equivalent
A1cso	Deduce the CVs (need not be shown explicitly) and state the (given) inequalities. There must be no errors in the working but the equation, if correct, can be solved easily so no working need be shown. Condone use of "and" instead of "or".
(c)	
M1	Differentiate the equation of C using the quotient rule. The denominator must be correct and the numerator must consist of the difference of 2 terms of the type shown. The product rule can be used.
A1	Either numerator term correct
A1	Fully correct numerator
ALT:	**Product Rule** $y=\left(2+4x-x^2\right)\left(2x+1\right)^{-1}$
M1A1 **A1**	$\frac{\mathrm{d}y}{\mathrm{d}x}=\left(4-2x\right)\left(2x+1\right)^{-1}-2\left(2+4x-x^2\right)\left(2x+1\right)^{-2}$ M1: rewrite without denominator and attempt product rule. Difference of 2 terms of the form shown needed (Difference because of the negative power) A1 Either term correct A1 Second term correct
M1	Equate the numerator of their derivative to 0 and solve to x = ...(any valid method of solving a quadratic with 2 or 3 terms) If product rule used must multiply through by $\left(2x+1\right)^2$
A1	Both x values correct
A1	Both stationary points correct
ALT	
M1	Use implicit differentiation on the re-arranged equation
A1	Correct derivative of $\left(2y-4\right)x$ (inc use of product rule)
A1	Fully correct derivative
M1	Set $\frac{\mathrm{d}y}{\mathrm{d}x}=0$ and use the result from (b) to obtain solutions
A1	One correct stationary point
A1	Both correct stationary points
(d)	
B1	Shape: Two parts, one above $y = 3$ and the other below $y = 2$
(i)M1	Attempt to find the x coordinates of the crossing points
A1	Correct coordinates shown on their sketch, 2 crossing points only. $y = 0$ need not be seen. There must be a curve through these points.
(ii)B1	The asymptote must be drawn and labelled (by its equation or by showing the x coordinate of the point where it crosses the x-axis). There must be at least one part of the curve which is asymptotic to the line $x=-\frac{1}{2}$. No part of the curve should touch/cross the asymptote or curve dramatically away from the line.
(iii)B1ft	Label the stationary points with their coords. Follow through provided the result is sensible.

Type MS · Finding stationary points

Answer 4 of 8 · Question 9 part c · 6 marks

2020 NOV · Paper 02 · extract 3 of 3 · find part c in this mark scheme extract

Question Number	Scheme	Marks

Type MS · Finding stationary points

Answer 5 of 8 · Question 11 part c · 7 marks

2024 JUN · Paper 02 (regional R) · extract 1 of 3 · find part c in this mark scheme extract

11(a)	$cos(A+A) = cos\,A\,cos\,A - sin\,A\,sin\,A$ $\Rightarrow cos\,2A = cos^2 A - (1 - cos^2 A)$ $cos\,2A = 2\,cos^2 A - 1 *$	M1 A1 cso [2]
(b)	$(2\cos^2 A - 1)^2 = (\cos 2A)^2$ $\cos 4A = 2\cos^2 2A - 1 \Rightarrow \cos^2 2A = \frac{\cos 4A + 1}{2}$ $(2\cos^2 2A - 1)^2 = \frac{\cos 4A + 1}{2} *$	M1 M1 A1 cso [3]
(b) **ALT**	$cos\,4A = 2\,cos^2 2A - 1$ $= 2(2\,cos^2 A - 1)^2 - 1$ $(2\,cos^2 A - 1)^2 = \frac{cos\,4A + 1}{2} *$	M1 M1 A1 cso [3]
(c)	$y = \frac{sin\,2x}{2} + \frac{(2\,cos^2 x - 1)^2}{2} + \frac{1}{8} \Rightarrow y = \frac{sin\,2x}{2} + \frac{cos\,4x}{4} + \frac{3}{8}$ $\frac{dy}{dx} = cos\,2x - sin\,4x$ $\frac{dy}{dx} = cos\,2x - 2\,sin\,2x\,cos\,2x \{= 0\}$ $\{cos\,2x(1 - 2\,sin\,2x) = 0\} \Rightarrow sin\,2x = \frac{1}{2}$ $\Rightarrow x = \frac{\sin^{-1}\left(\frac{1}{2}\right)}{2} = \cdots$ $x = \frac{\pi}{12}$ $y = \frac{sin2\left("\frac{\pi}{12}"\right)}{2} + \frac{cos4\left("\frac{\pi}{12}"\right)}{4} + \frac{3}{8} = \frac{3}{4} \{\Rightarrow \left(\frac{\pi}{12}, \frac{3}{4}\right)\}$	M1 dM1 B1 ddM A1 M1A1 [7]
		Total 12 marks

Type MS · Finding stationary points

Answer 5 of 8 · Question 11 part c · 7 marks

2024 JUN · Paper 02 (regional R) · extract 2 of 3 · find part c in this mark scheme extract

Part	Mark	Notes
(a)	M1	For correct use of $cos(A+B) = cos\,A\,cos\,B - sin\,A\,sin\,B \Rightarrow$ $cos(A+A) = cos\,A\,cos\,A - sin\,A\,sin\,A$ ***OR*** $cos\,2\,A = cos^2\,A - sin^2\,A$ **with** the Pythagorean identity to eliminate sine from the identity.
	A1 cso	Achieves printed answer with no errors seen. Left hand side must be cos2A not cos(A+A)
(b)	M1	Uses the given result from (a), replaces $(2\cos^2 A - 1)^2$ with $(\cos 2A)^2$
	M1	Uses the given result from (a) or a correct identity to replace $\cos^2 2A$ with $\frac{\cos 4A+1}{2}$
	A1 cso	Fully correct proof showing all necessary steps. (If they meet in the middle, they must have a conclusion, accept e.g. shown, proved, L=R, etc.)
(b) ALT	M1	Uses the given result from (a) on cos 4A
	M1	Uses the given result from (a) or a correct identity on cos 2A
	A1 cso	Fully correct proof showing all necessary steps. (If they meet in the middle, they must have a conclusion, accept e.g. shown, proved, L=R, etc.)
(c)	M1	Rearranges the given y into the form $\frac{\sin 2x}{2} + \frac{\cos 4x+1}{k} + l$, oe, where k and l are nonzero constants Or $\frac{\sin 2x}{2} + \frac{\cos 4x}{P} + Q$, oe, where P and Q are nonzero constants (Allow mixed variables x and A for this method mark only)
	dM1	For an attempt to differentiate their y into the form $P\cos(2x) \pm Q\sin(4x)$ *oe, where P, Q are nonzero constants* **Depends on the previous method mark.**
	B1	Uses $\sin 4x = 2\sin 2x\cos 2x$ or $\sin 4x = \sqrt{1-\cos^2 4x}$ oe in their dy/dx (may be implied by correct working)
	ddM1	Sets their differentiated expression = 0 (can be implied) and attempts a correct method to solve for x=... **Depends on both previous method marks.**
	A1	For $x = \frac{\pi}{12}$, ignore extra x values are outside of given range. Withhold A mark if extra solution seen in the given range.
	M1	For attempting to find the value of y using their x, found by solving their $\frac{dy}{dx} = 0$, provided x is in the range $0 \leqslant x \leqslant \frac{\pi}{6}$ Substitutes into the given y **or** their y, correct answer implies this mark, but for incorrect answer, must see explicit substitution
	A1	For the correct exact coordinates of P, $\left(\frac{\pi}{12}, \frac{3}{4}\right)$ *or* $\left(\frac{\pi}{12}, 0.75\right)$, accept $x = \frac{\pi}{12}$, $y = \frac{3}{4}$ *or* $x = \frac{\pi}{12}$, $y = 0.75$

Type MS · Finding stationary points

Answer 5 of 8 · Question 11 part c · 7 marks

2024 JUN · Paper 02 (regional R) · extract 3 of 3 · find part c in this mark scheme extract

Pearson Education Limited. Registered company number 872828
with its registered office at 80 Strand, London, WC2R 0RL, United Kingdom

Type MS · Finding stationary points

Answer 6 of 8 · Question 7 part b.i · 5 marks

2022 JUN · Paper 01 · extract 1 of 4 · find part b.i in this mark scheme extract

7 a (i)	f $x = \int (4x^3 - 12x^2 - 19x + 12)\,dx = \frac{4}{4}x^4 - \frac{12}{3}x^3 - \frac{19}{2}x^2 + 12x + D$ oe For the point $4, -104$ it follows that $-104 = (4)^4 - 4(4)^3 - \frac{19}{2}(4)^2 + 12(4) + D$ * $-104 = 256 - 256 - 152 + 48 + D \Leftrightarrow D = 0$*	M1 A1 M1 A1* cso (4)
a (ii)	$x = 0.5$ f $'(x) = 4(0.5)^3 - 12(0.5)^2 - 19(0.5) + 12 = 0$ f$''(x) = 12x^2 - 24x - 19$ $x = 0.5$ f$''(x) = 12(0.5)^2 - 24(0.5) - 19 (= -28) < 0$ Therefore maximum *	M1 M1 A1 cso (3)
b (i)	f$'(x) = (2x-1)(2x^2 - 5x - 12)$ f$'(x) = (2x-1)(2x+3)(x-4) \rightarrow x =$ $x = -\frac{3}{2}$ or $x = 4$, $\left[x = \frac{1}{2}\right]$ $A\left(-\frac{3}{2}, -\frac{333}{16}\right)$ $B(4, -104)$	M1 M1 A1 A1 A1 (5)
b (ii)	f$''(x) = 12(-1.5)^2 - 24(-1.5) - 19 (= 44) > 0$ Therefore minimum $x = 4$ f$''(x) = 12(4)^2 - 24(4) - 19 (= 77) > 0$ Therefore minimum (Read notes carefully for allocation of these marks) **Alternative** As we have a maximum at $x = 0.5$ and C is a continuous curve $x = -\frac{3}{2}$ is a minimum and $x = 4$ is a minimum	ddM1 A1 A1 (3) {ddM1} {A1}{A1} (3)
		Total 15 marks

Type MS · Finding stationary points

Answer 6 of 8 · Question 7 part b.i · 5 marks

2022 JUN · Paper 01 · extract 2 of 4 · find part b.i in this mark scheme extract

Part	Marks	Notes
(a) (i)	**M1**	Attempt to integrate – see general guidance for minimally acceptable attempt – at least one term must be fully correct in this integration (unsimplified).
	A1	Fully correct integration – does not need to include a constant of integration, need not be simplified.
	M1	Substitution of $x = 4$ and $y = -104$ into their expression, which must include a constant of integration.
	A1 cso	Obtains the given result, making it clear the constant of integration is 0 or listing fully the function at the end after fully correct working. There must be no incorrect working for this final mark to be awarded.
Note: candidates may assume $D = 0$ & show, by substituting, (4, -104) lies on the curve **final M1 A1**		
(a) (ii)	**M1**	Substitution of $x = 0.5$ into $f'(x)$ and shows $f'(x) = 0$. A candidate may also solve $f'(x) = 0$ and solve to get $x = 0.5$.
	M1	For finding $f''(x)$ - see general guidance for minimally acceptable attempt at differentiation. At least one term must be fully correct for this differentiation.
	A1 cso*	Substitutes $x = 0.5$ into a fully correct second derivative, shows that the result is negative and draws a conclusion. This can be as simple as "shown" or #. It is not necessary to work out the actual value of $f''(0.5)$ if clear substitution is shown, but if calculated, it must be correct. No incorrect work for this mark to be awarded. Note: as always, this A mark must follow 2 M1 marks.

Type MS · Finding stationary points

Answer 6 of 8 · Question 7 part b.i · 5 marks

2022 JUN · Paper 01 · extract 3 of 4 · find part b.i in this mark scheme extract

(b)	Part (i) and (ii) may be marked together	
(i)	**M1**	A clear attempt to divide by $(2x-1)$ or compare coefficients. The student must arrive at a quadratic factor of the form $(2x^2+Ax-12)$ if dividing It is also possible to divide by $\left(x-\frac{1}{2}\right)$ and arrive at a quadratic factor of the form $4x^2+Bx-24$
	M1	Uses a complete method to solve their 3TQ, see general guidance for a minimally acceptable attempt. Must progress to $x=$.
	A1	$x=-\frac{3}{2}$ or $x=4$ Award M1 M1 A1 if **both** correct values are given without working.
	A1	$A\left(-\frac{3}{2},-\frac{333}{16}\right)$ or $B(4,-104)$ Allow x = and y = (clearly paired) for **both** final A marks.
	A1	$A\left(-\frac{3}{2},-\frac{333}{16}\right)$ and $B(4,-104)$ accept 20.8 or better – answer is 20.8125
	Factorising in (a)(ii) and then using in this part can be awarded the marks above.	
(b) (ii)	**ddM1**	Correctly substitutes $x=-\frac{3}{2}$ or $x=4$ into their $f''(x)$. Their values Dependent on both previous M marks in part (b). A fully correct evaluation of the second derivative can imply this mark ie sight of 77 and 44. Part (i) and (ii) may be marked together
	A1	Substitutes $x=-1.5$ **or** $x=4$ into a fully correct second derivative, shows that the result is positive and draws the conclusion this is a minimum. It is not necessary to work out the actual value of $f''(x)$ if clear substitution is shown, but if calculated, it must be correct.
	A1	Substitutes $x=-1.5$ **and** $x=4$ into a fully correct second derivative, shows that the result is positive and draws the conclusion that both values of x give a minimum. It is not necessary to work out the actual value of $f''(x)$ if clear substitution is shown, but if calculated, it must be correct.
ALT	**ddM1**	For stating the curve is continuous and there is a maximum at $x=0.5$ Dep both previous M
	A1	For stating "as we have a maximum at $x=0.5$" and progressing to state either A or B must be a minimum.
	A1	Fully correct conclusion, stating A and B must both be minimum points.
There **MUST** be an appropriate justification for these marks to be awarded, such as that given in M1		

Type MS · Finding stationary points

Answer 6 of 8 · Question 7 part b.i · 5 marks

2022 JUN · Paper 01 · extract 4 of 4 · find part b.i in this mark scheme extract

Question number	Scheme	Marks
	4 … 3×500 …	M1 A1

Type MS · Finding stationary points

Answer 7 of 8 · Question 10 part b · 4 marks

2021 NOV · Paper 01 · extract 1 of 4 · find part b in this mark scheme extract

10 (a)	$\frac{1}{2}+\sin 3x=0 \Rightarrow \sin 3x=-\frac{1}{2} \Rightarrow 3x=-\frac{\pi}{6},\frac{7\pi}{6},\frac{11\pi}{6} \Rightarrow x=\frac{7\pi}{18},\frac{11\pi}{18}$ Coordinates of M are $\left(\frac{7\pi}{18},\ 0\right)$* Coordinates of N are $\left(\frac{11\pi}{18},\ 0\right)$ **ALT** $\frac{1}{2}+\sin\left(3\times\frac{7\pi}{18}\right)=0$ Coordinates of M are $\left(\frac{7\pi}{18},0\right)$ * Coordinates of N are $\left(\frac{11\pi}{18},0\right)$	M1 A1 A1 [3] [M1 A1 A1]
(b)	$\frac{dy}{dx}=3\cos 3x=0 \Rightarrow 3x=\frac{\pi}{2} \Rightarrow x=\frac{\pi}{6}$ $y=\frac{1}{2}+\sin 3\left(\frac{\pi}{6}\right)=1.5$ $\left[\text{Coordinates of point } A \text{ are } \left(\frac{\pi}{6},\ 1.5\right)\right]$ **ALT** Max of sine curve is 1 so that $\frac{1}{2}+1=\frac{3}{2} \Rightarrow y=\frac{3}{2}$ $\sin 3\theta = 1 \ \Rightarrow\ 3\theta=\frac{\pi}{2} \Rightarrow\ \theta=\frac{\pi}{6}$ $\left[\text{Coordinates of point } A \text{ are } \left(\frac{\pi}{6},\ \frac{3}{2}\right)\right]$	M1A1 dM1A1 [4] [M1A1 dM1A1]
(c)	Uses the given x coordinate for point M At point M $\frac{dy}{dx}=3\cos\left(3\times\frac{7\pi}{18}\right)=-\frac{3\sqrt{3}}{2}$ $y-0=-\frac{3\sqrt{3}}{2}\left(x-\frac{7\pi}{18}\right) \Rightarrow 12y+18\sqrt{3}x-7\sqrt{3}\pi=0$ o.e.	B1 M1A1A1 [4]
(d)	$A=\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2}+sin\,3x\right)\,dx+\left\|\left[\int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2}+sin\,3x\right)\,dx\right]\right\|$ $A=\left[\frac{1}{2}x-\frac{cos\,3x}{3}\right]_0^{\frac{7\pi}{18}}+\left\|\left[\frac{1}{2}x-\frac{cos\,3x}{3}\right]\right\|_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}$ $A=\left[\left(\frac{1}{2}\times\frac{7\pi}{18}-\frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right)-\left(0-\frac{cos\,3\times 0}{3}\right)\right]$	M1 M1 M1

Type MS · Finding stationary points

Answer 7 of 8 · Question 10 part b · 4 marks

2021 NOV · Paper 01 · extract 2 of 4 · find part b in this mark scheme extract

	$+\left\|\left[\left(\frac{1}{2}\times\frac{11\pi}{18}-\frac{cos\,3\left('\frac{11\pi}{18}'\right)}{3}\right)-\left(\frac{1}{2}\times\frac{7\pi}{18}-\frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right)\right]\right\|$ $A=[(1.23287)+(0.22828)]=1.46115\ldots\approx 1.46$	A1 [4]
		Total 15 marks

Type MS · Finding stationary points

Answer 7 of 8 · Question 10 part b · 4 marks

2021 NOV · Paper 01 · extract 3 of 4 · find part b in this mark scheme extract

	A1	For correctly obtaining $x = \frac{\pi}{6}$
(c)	**B1**	For correct gradient at point M $\left(\frac{dy}{dx}\right) = -\frac{3\sqrt{3}}{2}$
	M1	For a fully correct method of finding the equation of the tangent to C at M $y - 0 = ' -\frac{3\sqrt{3}}{2} ' \left(x - \frac{7\pi}{18}\right)$ Allow use of their $-\frac{3\sqrt{3}}{2}$ obtained from substitution of $x = \frac{7\pi}{18}$ into their $\frac{dy}{dx}$.
	A1	For a correct equation of the tangent in any form: $y - 0 = -\frac{3\sqrt{3}}{2}\left(x - \frac{7\pi}{18}\right)$
	A1	For correct equation of the tangent in the required form: $12y + 18\sqrt{3}x - 7\sqrt{3}\pi = 0$ Accept integer multiples of this equation.
(d)	**M1**	For identifying the correct limits to find the area above the curve and the area below the curve: $\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2} + \sin 3x\right) + \left|\int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2} + \sin 3x\right)\right|$ or $\int_0^{\frac{7\pi}{18}}\left(\frac{1}{2} + \sin 3x\right) - \int_{\frac{7\pi}{18}}^{\frac{11\pi}{18}}\left(\frac{1}{2} + \sin 3x\right)$ Allow use of their $\frac{11\pi}{18}$. Needs to correctly indicate dealing with areas above and below axis.
	M1	For an attempt to integrate $\frac{1}{2} + \sin 3x$ obtaining: $\frac{1}{2}x - k\cos 3x$ where $k \neq -3$ For this mark ignore incorrect / absent limits.
	M1	For substituting limits correctly into their integrated expression (must be a changed expression). $A = \left[\left(\frac{1}{2} \times \frac{7\pi}{18} - \frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right) - \left(0 - \frac{cos\,3 \times 0}{3}\right)\right]$ $+ \left\|\left[\left(\frac{1}{2} \times \frac{11\pi}{18} - \frac{cos\,3\left(\frac{'11\pi'}{18}\right)}{3}\right) - \left(\frac{1}{2} \times \frac{7\pi}{18} - \frac{cos\,3\left(\frac{7\pi}{18}\right)}{3}\right)\right]\right\|$ Allow use of their $\frac{11\pi}{18}$. Must show substitution or M0.
	A1	For correctly obtaining awrt 1.46

Type MS · Finding stationary points

Answer 7 of 8 · Question 10 part b · 4 marks

2021 NOV · Paper 01 · extract 4 of 4 · find part b in this mark scheme extract

Question number	Scheme	Marks

Type MS · Finding stationary points

Answer 8 of 8 · Question 4 part a · 4 marks

2020 NOV · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Question number		
4 (a)	$\frac{dy}{dx} = 3x^2 - 6x - 24$	M1
	$"(x+2)(x-4)" = 0$	M1
	$x = -2$ or $x = 4$	A1
	(–2, 34) and (4, –74)	A1 (4)
(b)	$\frac{d^2y}{dx^2} = "6x - 6"$ and substitution of their –2 or 4 or consider values of $\frac{dy}{dx}$ either side of their –2 or 4 or use properties of a cubic*	M1
	(–2, 34) maximum and (4, –74) minimum	A1cso (2) **[6]**

Part	Mark	Additional Guidance
(a)	M1	General guidance – an attempt to differentiate, power of at least one term must decrease by one. Also, no power must increase.
	M1	Equates their derivative = 0 and attempts to solve by any method – **see general guidance.**
	A1	Both correct values for x
	A1	All values correct, listed as coordinates or correctly paired $x = \ldots\ldots$ $y = \ldots\ldots$
(b)	M1	**Correctly differentiates their derivative** from part a and substitutes one of their x values. Allow sight of 18 or -18 following a correct differentiation to imply this mark.
	A1 cso	Allow this mark if there is an incorrect y value from part a Must correctly make the argument they've chosen and identify the points as shown. If substituting x = -2 and x = 4 or values either side of the these, their evaluations of the substitutions must be correct. * A convincing argument about the shape of a positive cubic curve and position of the minimum and maximum point must be made. Send to review, if in doubt. In the conclusion, it must be clearly stated which coordinate or value of x is a maximum and which is a minimum. Just a substitution of a value and stated max or min is insufficient.

Type MS · Finding stationary points

Answer 8 of 8 · Question 4 part a · 4 marks

2020 NOV · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks